Fire Identification: A Satellite Data Analysis of the 2019–2020 Fire Season in Australia

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***Abstract* — This project set out to determine whether satellite data from NASA collected between 2019-2020 is suitable for predicting if an observation indicates a real high threat fire. Methods for both linear and categorical response variables were applied to the data sets to determine the best option for reasonable predictions. Results indicate that the satellite data can be used for predicting high threat fires with accuracies reaching 82%. Further work would likely see improvements in prediction through additional feature data collection for application in advanced models with extensive hyperparameter tuning.**

I. INTRODUCTION

Wildfires are among the most destructive natural hazards on earth, and few recent events illustrate their scale as starkly as Australia's 2019–2020 bushfire season. Widely known as the "Black Summer," the season burned an estimated 24 million hectares. It was one of the most intense and catastrophic fire seasons on record in Australia. The wildfire caused 33 direct deaths, destroyed more than 3,000 structures, and displaced or killed an estimated three billion animals. Responding to events of this magnitude depends on detecting active fires rapidly and distinguishing genuine, high-threat fires from false alarms, permitting the limited firefighting resources to be directed where they are most needed. Satellite remote sensing is central to this task. The Moderate Resolution Imaging Spectroradiometer (MODIS), aboard NASA's Terra and Aqua satellites, detects thermal anomalies in the mid-infrared bands and reports candidate active-fire pixels through the standard MODIS MOD14/MYD14 Fire and Thermal Anomalies product, which are distributed in near real time by NASA's Fire Information for Resource Management System (FIRMS). Each detection carries a detection confidence value; for MODIS this ranges from 0% to 100% and is used to assign one of three fire classes — low-, nominal-, or high-confidence fire. The confidence field was added to help users gauge the quality of individual fire pixels, and users whose goal is to reduce false alarms may retain only nominal- and high-confidence pixels and treat low-confidence pixels as non-fire. This matters because the near-real-time product is not considered science quality, making the confidence field an important safeguard against errors of commission.

This project uses the MODIS confidence field as the prediction target for Australian fire detections from the 2019–2020 season. The central question is whether the remaining satellite measurements — brightness, fire radiative power, brightness temperature, scan geometry, acquisition time, location, and platform — can predict whether an observation is a high-confidence fire. The reduction of false alarms has direct operational value which means fewer wasted

dispatches, lower cost and free funding for equipment, prevention, and staffing during peak seasons of wildfires.

To approach the problem from multiple angles, we frame confidence both as a continuous response (regression) and as a binary high/low label (classification) and apply a range of methods: logistic and linear regression baselines, regularized regression via elastic net, generalized linear models with distribution-appropriate targets, principal component analysis for dimension reduction, and K-Means clustering. Each model is evaluated with metrics suited to its task, including accuracy, precision, recall, ROC-AUC, log-loss, RMSE, and R^2. The remainder of the paper describes the methods, data exploration, and results of each analysis.

II. METHODS

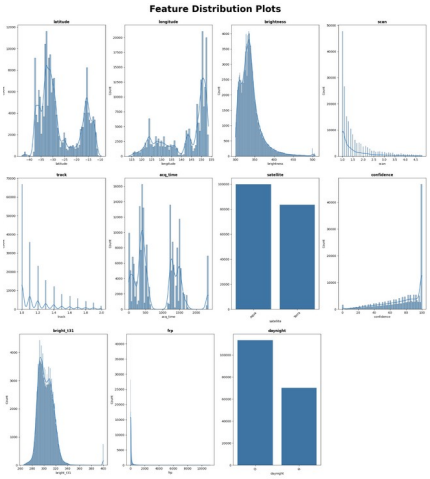
We used a logistic regression model as the baseline model. This is a simple model that provided us with something to compare other models to. It is suitable for handling a large data set with over 100,000 observations. The model is resistant to overfitting and is good for initial model interpretability. Subsequent models used for experimentation are Elastic Net Regression, generalized linear models (GLMs), Principal Component Analysis (PCA), and K-Means clustering. To evaluate the classification models, we used the following statistics: accuracy, precision and recall, log-loss, and ROC/AUC; the regression models were evaluated with MAE, RMSE, and R^2. Combining the information from these statistics provided us with a comprehensive look at how well the models perform.

III. DATA EXPLORATION AND VISUALIZATION

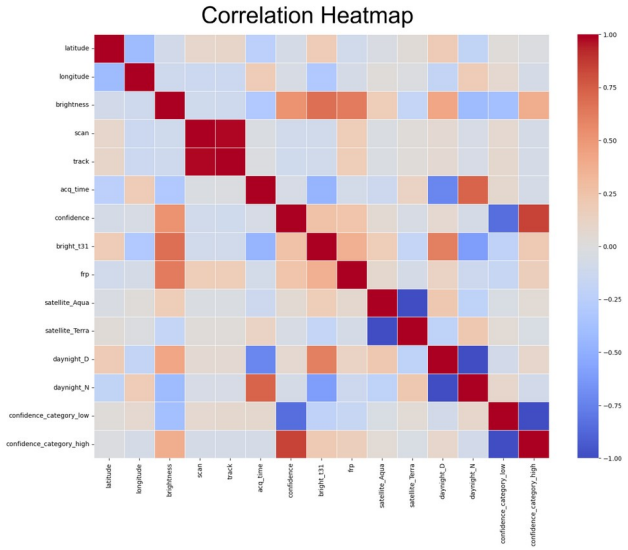
Data Set:
https://www.kaggle.com/datasets/carlosparadis/fires-from-space-australia-and-new-zeland?select=fire_nrt_M6_96619.csv

Rows: 183,593
Columns: 14
Feature Types:
- Categorical Nominal
- Numerical Continuous
- Date-Time

Data Split:
- Training 60% 110155
- Validation 20% 36719
- Testing 20% 36719



(Fig 1.) Feature Distribution Plots - The histograms and bar plots are the feature distribution plots for all of the features in the dataset. We can see that some variables have high skew.



(Fig 2.) Correlation Heatmap - A heatmap of all features except for the date-time feature acq_date were plotted after one-hot encoding to check for collinearity issues.

A. Transformations

Our response variable is confidence, that is the confidence in percentage that the observation is a fire. Since the project calls for categorical and continuous response variable analysis, the confidence feature was used in linear regression and we engineered categorical features by splitting confidence into two categories, low and high, for logistic regression. All values below 60% are considered low and values above 60% are considered high. During EDA a missingness report was run to search for any feature with missing data in its observations. No missing values were found so there was no need for replacement or imputation. Histograms were created for all the features to provide insight into their distributions. We noticed that the histograms showed heavy skewness for some features. To address this, we performed scaling on all the numeric columns. The data also contains features with categorical values. For these features we performed one-hot encoding to ensure that the simpler models could process the data. One-hot encoding was chosen to ensure that the categories could be differentiated, and one category would not be considered greater in value than another. A heatmap was created to check for high collinearity. We found nearly perfect collinearity for the features scan and track. Code was written to drop at least one feature if there were two or more features found to have 90% or greater collinearity. The result was that the feature called track was dropped.

IV. EXPLANATORY DATA ANALYSIS

A. Feature Selections

For both the logistic and linear regression baseline models, feature selection was performed using the exhaustive feature function from the mlxtend library. Since the data does not have many features, we did not put limitations on the feature combinations. We did specify that the model runs should use 5-fold cross validation. To determine the best feature subset the metrics chosen for comparison for logistic and linear regression are accuracy and R^2 respectively. In both cases the best subset selection results were the same and only contained one feature, brightness. This feature subset was compared to the feature importance plots for the models. The findings revealed that for the logistic regression model the frp feature was most important followed by brightness.

B. Regression and Classification Analysis

For the Logistic Regression model, the response variable is confidence_category_high, which is a binary variable with values of true or false. To

evaluate this model, we reviewed the confusion matrix, precision, recall and F1 scores, the ROC/AUC score, accuracy, log-loss and overall model training time. We can see by the confusion matrix where true positives and true negatives add up to 30126 of the 36719 validation observations in conjunction with the accuracy score of about 82% that there is room for improvement. We can see by the support numbers that there is a class imbalance in the dataset. There are nearly three times more true observations than false ones in the validation dataset. The precision, recall and F1 scores support that the model does a pretty good job of identifying true cases and a moderate to poor at identifying false cases. The ROC/AUC score of 87.69% suggests that the model has good discriminatory power. The log-loss of 0.38847 supports that the model is performing better than just guessing and heavily penalizes for confident incorrect predictions. Model training time is about 0.33 seconds.

For the Linear Regression model, the response variable is confidence, which is a continuous variable with values from 0 to 100. To evaluate this model we reviewed the MAE, MSE, RMSE and R^2 values as well as the overall model training time. The model's results produced an RMSE of 0.81498 which is higher than the MAE of 0.64797 indicating that the model is making some large errors. The R^2 value is 0.33048 meaning that only about 33% of the variance can be explained by this model. The overall model training time is 0.13 seconds.

C. Lasso and Ridge Via Elastic Net Regression

Linear regression estimates its coefficients by minimizing the squared error alone, which can produce unstable, high variance estimates when predictors are numerous or correlated and offers no mechanism to suppress uninformative features. Regularization addresses this by adding a penalty on coefficient magnitude to the loss, by trading a small increase in bias for a reduction in variance. Two penalties are standard in regularization: the ridge (L2) penalty, which shrinks all coefficients smoothly toward zero but never to exactly zero, and the lasso (L1) penalty, whose geometry drives some coefficients to exactly zero and automatically performs an implicit feature selection. Elastic net combines both penalties, which lets it shrink correlated predictors together like ridge while retaining the sparsity of lasso.

For standardized inputs, the elastic net estimates the coefficient vector β that minimizes the penalized objective

$$\min_{\beta} \frac{1}{2n}\|y - X\beta\|_2^2 + \alpha\rho\|\beta\|_1 + \frac{\alpha(1-\rho)}{2}\|\beta\|_2^2$$

where *n* is the number of training observations, α is the overall penalty strength, and ρ is the mixing parameter l1_ratio. Setting ρ=1 recovers pure lasso and ρ=0 recovers pure ridge, so that a single search over ρ spans both the ridge and lasso methods.

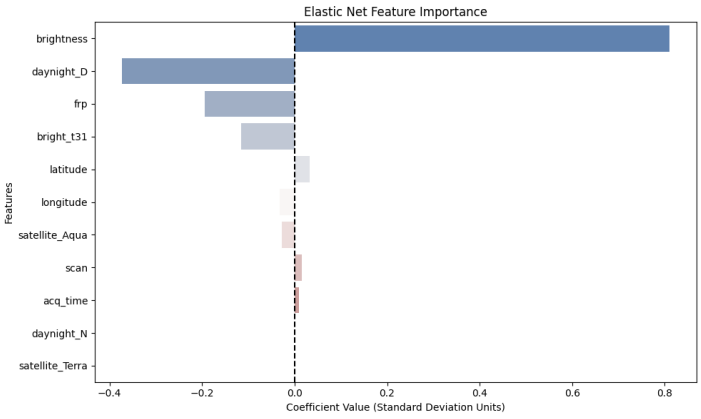
We fit the model with ElasticNetCV, which uses 5-fold cross-validation to select α and ρ jointly from a grid of l1_ratio values (0.1 through 1.0) and a path of 100 automatically generated α values. The response was the scaled continuous confidence, and all predictors were

standardized during preprocessing, which is required for the penalty to treat every coefficient on a common scale. Fitting on the scaled response also keeps the results directly comparable to the linear regression baseline.

The cross-validation selected an l1_ratio of 1.0 — the pure-lasso end of the spectrum — with a very small penalty of α=0.000526, and produced a validation MAE of 0.64808, RMSE of 0.81498, and R^2 of 0.33048. These values are effectively identical to the linear regression baseline. The coefficient estimates explain why: brightness dominated with a standardized coefficient near 0.81, followed at much smaller magnitudes by daynight_D, frp, and bright_t31, and no coefficient was meaningfully driven to zero. The two coefficients that appear as essentially zero (satellite_Terra and daynight_N) are numerical artifacts rather than lasso selection: each is the perfect complement of its one-hot partner, so once satellite_Aqua and daynight_D are in the model, their mirrors contribute nothing.

The interpretation is that the penalty had almost nothing to remove. Because the near-perfectly collinear feature track was already dropped in preprocessing, the remaining predictor set carries little redundancy, and the cross-validation responded by choosing a negligible penalty — reducing the elastic net to ordinary least squares in all but name. Elastic net therefore functions here as a regularization check that confirms the linear baseline rather than improving on it. It also reinforces a broader finding: the weak linear fit ($R^2 \approx 0.33$) reflects a genuinely limited linear relationship between the predictors and confidence, one that regularization cannot repair and that motivates the next step: choosing a target distribution that better matches the response.

(Fig 3.) Standardized coefficient estimates from the cross-validated elastic net on the scaled confidence response. brightness dominates; all other coefficients are small, and none is meaningfully shrunk to zero, indicating the selected penalty was near zero. The two coefficients at approximately zero (satellite_Terra, daynight_N) are the complementary members of one-hot pairs, not lasso-selected features.



D. GLMs With Appropriate Target Distributions

A generalized linear model extends ordinary regression through three components: a random component specifying the probability distribution of the response, a systematic component consisting of the linear predictor $\eta_i = \beta_0 + \sum_j \beta_j x_{ij}$, and a link function that connects the expected response to the linear predictor via $g(\mu_i) = \eta_i$, where $\mu_i = E[y_i]$. Choosing the distribution and link to match the response is what makes a GLM "appropriate": the distribution should reflect the response's support, which is its allowed

range, and its mean–variance relationship. We fit two GLM families, each matched to a differently shaped response.

For the binary response confidence_category_high, the natural choice is the Binomial (Bernoulli) family with the logit link, which is the GLM formulation of logistic regression:

$$\text{logit}(\mu_i) = \ln\left(\frac{\mu_i}{1-\mu_i}\right) = \beta_0 + \sum_j \beta_j x_{ij}$$

Here $\mu_i = E[y_i] = P$(high confidence), and the Binomial variance $\mu_i(1-\mu_i)$ties the spread of the response to its mean, which is correct for a probability bounded on [0, 1]. This model reached 82.03% accuracy, an ROC-AUC of 87.70%, and a log-loss of 0.38849. Beyond these classification metrics, the GLM framing supplies likelihood-based diagnostics: an AIC of 84269.22 and a residual deviance that falls from a null deviance of 127132.18 to 84249.22. The deviance drop is the GLM analogue of explained variation — the predictors reduce the model's lack-of-fit substantially relative to an intercept-only model — confirming that the features carry real signal. The confusion matrix (24,523 true positives, 5,599 true negatives, 4,086 false positives, 2,511 false negatives) shows the model is far stronger at identifying the majority "high" class than the minority "low" class, consistent with the roughly 3:1 class imbalance in the data.

For the continuous confidence response, which is a percentage bounded on [0, 100], we compared two distributions. The first is the Gaussian family with the identity link, which is ordinary linear regression expressed as a GLM:

$$\mu_i = E[y_i] = \beta_0 + \sum_j \beta_j x_{ij}, \qquad y_i \sim \mathcal{N}(\mu_i, \sigma^2)$$

The identity link places no bound on the fitted mean and assumes constant variance, so the model can predict impossible values below 0 or above 100 and does not reflect the skew of the data. The second treats confidence as a proportion, $p_i = \text{confidence}_i/100 \in [0,1]$, and models it with the same Binomial-logit form above, now with $\mu_i = E[p_i]$. Modeling the proportion respects the [0, 100] bound after back-transformation and matches the mean–variance behavior of percentage data. On the validation set the Gaussian model gave an RMSE of 20.41 and R^2 of 0.3305, while the proportion-based Binomial model gave an RMSE of 16.67 and R^2 of 0.5534 — a substantial improvement from simply choosing a distribution suited to the response. We did not use a Poisson or Gamma family here: Poisson is intended for unbounded counts, and Gamma requires strictly positive values whereas confidence can equal zero, so neither matches a bounded percentage as cleanly as the proportion-Binomial. Together, the two comparisons demonstrate the core lesson of GLMs — that aligning the target distribution with the nature of the response materially improves the fit.

Confusion Matrix Binomial GLM

Classification	Count
True Positives	24523
False Positives	4086

True Negatives	5599
False Negatives	2511

(Table 1) Classification outcomes for the Binomial (logit) GLM on the binary high-confidence label at a 0.5 probability threshold.

Binomial GLM Model Report

Metric	Value
Training Time	1.04 s
Accuracy	82.03 %
ROC-AUC score	87.70 %
Log Loss	0.38849
AIC	84269.22
NULL Deviance	127132.18
Residual Deviance	84249.22

(Table 2) Validation classification metrics and GLM likelihood diagnostics (AIC, null and residual deviance) for the Binomial-logit model.

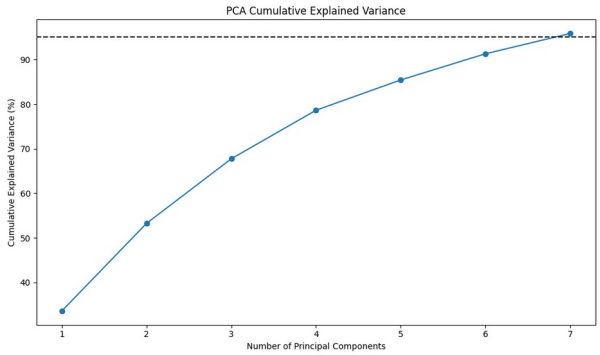
GLM Target Distribution Comparison (Continuous Confidence)

GLM Family (Link)	RMSE	R^2	Train Time
Gaussian (identity)	20.4088	0.3305	0.33 s
Binomial (logit) on proportion	16.6689	0.5534	0.83 s

(Table 3) Validation RMSE and R^2 for a Gaussian (identity) GLM versus a Binomial (logit) GLM fit to the proportion confidence/100. The proportion model fits better and cannot predict outside the valid [0, 100] range.

E. Dimension Reduction

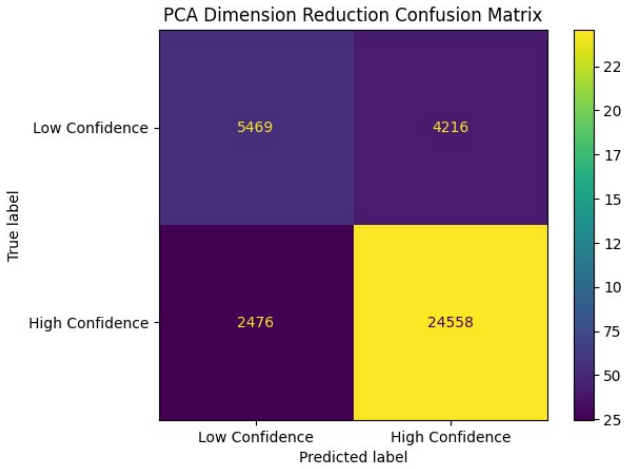
For dimension reduction, Principal Component Analysis, or PCA, was used. PCA was chosen because the dataset contains multiple numerical and encoded categorical predictors, and the goal was to reduce the number of predictors while keeping most of the useful information in the data. The PCA model was set to keep enough principal components to explain at least 95% of the variance. After fitting PCA on the training predictors, the original 11 predictors were reduced to 7 principal components. These 7 components explained about 95.83% of the total variance in the training data.



(Fig 4.) PCA Cumulative Explained Variance Plot - This plot shows how much total variance is explained as more principal components are added. The horizontal reference line marks 95% explained variance. The plot shows that 7 principal components were enough to keep most of the information from the original predictor set.

After dimension reduction, a logistic regression model was trained using the PCA components instead of the original predictors. The response variable was confidence_category_high, which identifies whether the fire confidence value was high or low. The PCA logistic regression model had a validation accuracy of 81.78% and a ROC-AUC score of 87.48%. This means that even after reducing the number of features, the model kept strong predictive

ability. The results suggest that PCA successfully reduced the dataset while preserving most of the information needed for classification.

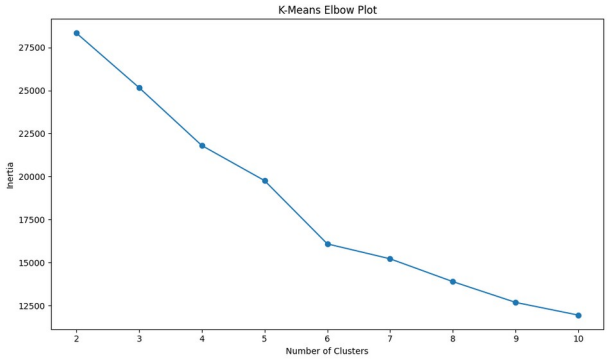


(Fig 5.) PCA Logistic Regression Confusion Matrix - This confusion matrix shows the number of correct and incorrect classifications for the low confidence and high confidence categories after using the PCA-reduced predictors.

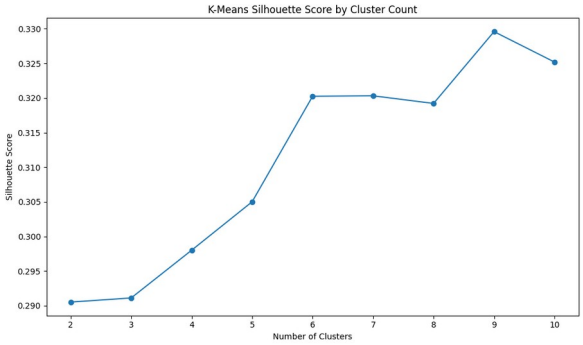
F. Clustering

For clustering, K-Means clustering was used to identify possible hidden groups in the fire observation data. Since clustering is an unsupervised method, the confidence category was not used to create the clusters. The clustering model was only fit using the predictor variables. PCA was also applied before clustering to reduce noise and make the clustering process more efficient. The PCA model again kept enough components to explain at least 95% of the variance.

To choose the number of clusters, different k values were tested and compared using inertia and silhouette score. The inertia values were used for the elbow plot, while the silhouette scores were used to evaluate how well-separated the clusters were. The best number of clusters selected by silhouette score was 9. The final K-Means model was then fit using 9 clusters. The training silhouette score was 0.3038, and the validation silhouette score was 0.3198.



(Fig 6.) K-Means Elbow Plot - This plot shows the inertia values for different numbers of clusters. The elbow plot helps show where adding more clusters begins to give smaller improvements.



(Fig 7.) K-Means Silhouette Score Plot - This plot compares silhouette scores for different values of k. The highest silhouette score was used to select the best number of clusters.

The silhouette scores show that there is some cluster structure in the data, but the clusters are not extremely separated. This makes sense because fire observations may overlap across features such as brightness, scan, frp, latitude, longitude, and acquisition time. After clustering, the cluster labels were compared with the actual confidence category to see whether the clusters had any relationship with low or high confidence fire

observations. This comparison was only done after clustering and was not used to train the K-Means model.

V. CONCLUSION

Overall, this project showed that satellite fire detection data can be used to analyze and predict fire confidence levels during the 2019–2020 fire season in Australia. The dataset contained 183,593 observations and 14 original columns, which were split into 60% training, 20% validation, and 20% testing sets. After preprocessing, including scaling, one-hot encoding, and removing the highly correlated feature track, several statistical and machine learning methods were applied. Across multiple models, brightness and frp were consistently important predictors, which makes sense because they directly relate to the strength and intensity of the detected fire signal.

The classification models showed solid performance for separating low and high confidence fire observations. The Binomial GLM reached about 82.03% accuracy, an ROC-AUC of 87.70%, and a log-loss of 0.38849. For the continuous confidence response, the proportion-based Binomial GLM performed better than the Gaussian GLM, with an RMSE of 16.67 and an R^2 of 0.5534 compared to the Gaussian model's RMSE of 20.41 and R^2 of 0.3305. PCA also worked well for simplifying the data, reducing the predictors from 11 features to 7 principal components while still explaining 95.83% of the variance. The PCA logistic regression model still produced an accuracy of 81.78% and an ROC-AUC of 87.48%, showing that most of the predictive information was preserved.

K-Means clustering was useful for exploring hidden groupings in the fire observations. The best number of clusters selected was 9, with a training silhouette score of 0.3038 and a validation silhouette score of 0.3198. These scores suggest that the data has some cluster structure, but the groups are not strongly separated. Overall, the analysis shows that the dataset contains strong predictive signal for fire confidence, especially through brightness-related variables, but performance could likely be improved further by adding more environmental, weather, or spatial features and by performing more advanced model tuning.

VI. EXECUTIVE SUMMARY

A. Overview & Mission

Fire departments play a key role in safety and disaster prevention. Employees work extensive shifts and experience high levels of stress and physical exertion when responding to calls. Equipment is expensive to obtain, requires maintenance and is highly specialized. Deploying a unit can cost the state hundreds or even thousands of dollars per deployment regardless of if an event turns out to be a real high threat situation or not. Our mission is to reduce the number of fire response unit deployments for false alarms by leveraging satellite data and prediction models.

B. Opportunity & Value Proposition

Being able to predict if a satellite observation is an actual high-threat bushfire or not can help the fire departments in Australia better allocate resources. If the number of false alarms can be reduced then

cost for sending out units to address them will also decrease, possibly leaving funding for better equipment, fire prevention tasks, or increased staffing to handle high occurrence seasons. Local states in Australia stand to benefit by a factor of around $1,000 per false alarm avoided. Other monetary gains can be realized through the reduction in health cost for employees dealing with the high levels of mental and physical demand that comes with the occupational choice of working in emergency response.

C. Performance & Growth

The models built to make predictions based on the NASA satellite data collected during the 2019-2020 fire season for Australia demonstrated good performance, achieving accuracies of up to 82%. Driven by data analysis and experimental results, we have established that there is high potential for reducing overall cost burden felt by states regularly responding to fire related emergency calls. Prediction model deployment should start with departments experiencing the highest call levels for fire related emergencies. Further deployments should be made as models improve and data related to the proof of concept from initial deployments are analyzed.

D. Looking Forward

It is our recommendation that states invest in more extensive data collection directed at filling out the feature space that is related to potential factors that increase the likelihood of a fire. If this data is then provided to data analysts for use in building predictive models, better results will follow. Making this short-term investment will produce long term monetary benefits for local fire departments, individual states and Australia as a whole.

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